

APRS

Theory and Operations

A Primer for Amateur Radio Operators

VHF APRS · HF APRS · Packet Structure · Practical Operation

*Understanding how the Automatic Packet Reporting System works —
from the radio signal to the map display.*

Part of the APRS Command documentation suite

APRS Command

github.com/KE4CON/APRS-Command

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What this primer covers

- What APRS is and how it works
- VHF APRS — frequencies, paths, and digipeaters
- Packet structure and data types
- APRS-IS — the internet backbone
- HF APRS — long-distance packet radio
- Practical operating guidance

Built on the vision of Bob Bruninga WB4APR · In memory of Roger Barker G4IDE

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Chapter 1

What Is APRS?

1.1 The Origin of APRS

Bob Bruninga WB4APR, a senior research engineer at the United States Naval Academy, implemented the earliest ancestor of APRS on an Apple II computer in 1982. This early version was used to map high frequency Navy position reports.

The first use of APRS was in 1984, when Bruninga developed a more advanced version on a VIC-20 for reporting the position and status of horses in a 100-mile (160 km) endurance run.

Over the next two years, Bruninga continued to develop the system, which he called the Connectionless Emergency Traffic System (CETS). Following a series of Federal Emergency Management Agency (FEMA) exercises using CETS, the system was ported to the IBM Personal Computer. During the early 1990s, CETS — then known as the Automatic Position Reporting System — continued to evolve into its current form.

As GPS technology became more widely available, “Position” was replaced with “Packet” to better describe the more generic capabilities of the system and to emphasize its uses beyond mere position reporting. Bruninga also noted that APRS was not meant to be a vehicle position tracking system, and can be interpreted as “Automatic Presence Reporting System.”

Bob continued developing and advocating for APRS until his passing in February 2022. His website at aprs.org remains the canonical reference for APRS theory and protocol. This primer draws heavily on his work and philosophy.

1.2 Situational Awareness, Not Just Tracking

Bob consistently pushed back against describing APRS as a “tracking system.” Tracking implies following a moving object. APRS is broader: it is a situational awareness tool. When you look at an APRS map, you see not just where stations are, but what they are doing.

A well-deployed APRS network during an emergency activation shows:

- Field operators moving to their assigned positions
- Weather stations reporting real-time conditions across the affected area
- Net control stations transmitting their location and frequency
- Objects marking shelter locations, staging areas, and points of interest
- Messages passing between operators in the field and command

This is a common operating picture — the same concept used by the Incident Command System. APRS gives amateur radio operators that capability using nothing but radios and a shared digital protocol.

 Tip

When you deploy APRS for an event or activation, think about what information other operators need to see on their map — not just your location. Objects, status messages, and weather data all contribute to the picture.

1.3 How APRS Fits Into Amateur Radio

APRS sits within the broader world of amateur radio digital modes — communications that use computers or digital encoders to send structured data over radio. Unlike modes such as FT8 (optimized for weak-signal contacts) or Winlink (email over radio), APRS is designed for real-time tactical information sharing on a shared channel.

APRS uses the AX.25 packet radio protocol, which is the amateur radio adaptation of the X.25 networking standard. Every APRS packet is an AX.25 frame containing an origin callsign, a destination, optional digipeater addresses, and the data payload. On VHF, this is transmitted as audio frequency shift keying (AFSK) — two audio tones representing the ones and zeros of the digital signal — into a standard FM radio.

This means APRS works with ordinary VHF FM transceivers. You do not need a special radio. You need a way to generate and decode the audio tones: either a hardware TNC (Terminal Node Controller), a software modem running on a computer with an audio interface, or a radio with a built-in TNC.

Note

APRS requires an amateur radio license to transmit. In the United States, a Technician class license or higher is required. You can receive and view APRS data without a license using APRS-IS (the internet gateway network) or by listening passively on 144.390 MHz.

1.4 What APRS Is Not

Understanding what APRS is designed for helps set appropriate expectations for what it can and cannot do well.

APRS is...	APRS is not...
A real-time situational awareness tool	A reliable point-to-point communication system
A shared broadcast medium	A private or encrypted channel
Best for one-to-many information sharing	Ideal for high-volume two-way conversations
Designed for tactical operational data	A replacement for voice radio in emergencies
A snapshot of the current operating picture	A complete historical log (packets can be lost)
A system built on best-effort delivery	A guaranteed delivery mechanism

Packets on the APRS network are not acknowledged at the RF level (with the exception of APRS messages, which have their own acknowledgment mechanism). A digipeater relays what it hears; a receiver records what it receives. Neither guarantees that any specific station received your transmission. This is by design — the shared broadcast model is what makes APRS scalable to thousands of simultaneous stations without coordination.

Chapter 2

The RF Layer: VHF APRS

2.1 The Shared Channel

In North America, VHF APRS operates on a single shared frequency: 144.390 MHz. Every station in range — your handheld, mobile radio, home station, nearby digipeater, and every other APRS station within line of sight — all transmit on this same channel without coordinating with each other. This is the APRS channel. When your radio is on 144.390 MHz, you are on the APRS channel.

The shared nature of the channel is fundamental to how APRS works and also its primary limitation. Because everyone shares one frequency, packet collisions are common. When two stations transmit simultaneously, both packets are corrupted and neither is received correctly. The network is designed to tolerate this — stations retransmit regularly, so a missed packet is usually recovered on the next beacon cycle.

 Important

The APRS channel is a broadcast medium, not a conversation channel. Transmitting continuously or with very short beacon intervals clogs the channel for everyone in your area. Section 8.2 covers beacon interval best practices.

2.2 How Packets Travel Over Radio

When your station transmits an APRS beacon, the following sequence occurs:

- Your TNC or software modem encodes the packet data as audio tones — 1200 Hz for a "mark" (binary 1) and 2200 Hz for a "space" (binary 0), transmitted at 1200 bits per second.
- These tones are fed into your radio's audio input. The radio transmits them as an FM signal on 144.390 MHz.
- Every station within radio range receives the FM signal and decodes the audio tones back to digital data.
- If the packet is addressed with a digipeater path (such as WIDE1-1,WIDE2-1), nearby digipeaters that receive it will re-transmit it, extending the range.
- Stations with internet connections (iGates) forward received packets to the global APRS-IS network.

A typical VHF APRS packet travels 15–50 miles from a mobile station with a good antenna, farther from a hilltop or well-located home station, and potentially hundreds of miles with digipeater relaying.

2.3 Modulation: AFSK and AX.25

VHF APRS uses Bell 202 AFSK (Audio Frequency Shift Keying) modulation. The two tones — 1200 Hz (mark) and 2200 Hz (space) — are transmitted at 1200 baud. This is a relatively slow data rate by modern standards, but it is well-matched to the characteristics of FM radio and the available bandwidth.

The digital protocol underlying APRS packets is AX.25 — the amateur radio adaptation of the X.25 packet switching protocol. An AX.25 frame contains:

- Flag bytes — synchronization markers at the start and end of the frame

- Destination field — the APRS software identifier (e.g., APRS, APNU, etc.)
- Source field — the transmitting station's callsign and SSID
- Digipeater fields — up to 8 relay station addresses (the "path")
- Control and PID bytes — protocol identifiers
- Information field — the APRS data payload (position, message, weather, etc.)
- FCS — a 16-bit frame check sequence for error detection

Note

The 1200 baud rate of VHF APRS dates to the early days of the protocol and the limitations of 1990s hardware. Various proposals for higher-speed APRS alternatives have been made over the years, but 1200 baud remains the standard for interoperability. The entire network depends on all stations being able to receive each other.

2.4 The TNC: Hardware and Software

A TNC (Terminal Node Controller) is the device that sits between your radio and your computer, converting between digital data and the audio tones used to carry that data over radio. In the early days of packet radio, every station needed a dedicated hardware TNC. Today, software modems running on a computer can perform the same function using a sound card or audio interface connected to the radio.

Type	Examples	Best For
Hardware TNC	Kantronics KPC-3+, TNC-Pi, Mobilinkd TNC4, DigiRig	Reliable stand-alone operation; field use; digipeater builds
Software modem (sound card)	Direwolf, UZ7HO Soundmodem	Flexibility; modern computers; iGate builds
Radio with built-in TNC	Kenwood TM-D710, TM-V71, TH-D74	All-in-one mobile/portable; no computer required for basic operation
Sound card interface	Signalink USB, DigiRig, RigBlaster	Bridge between software modem and radio audio/PTT

The choice of TNC or software modem does not affect how your packets are received by others on the network — the output is the same Bell 202 AFSK audio. It affects reliability, portability, and the complexity of your setup.

Tip

For a home iGate or digipeater, Direwolf running on a Raspberry Pi with a sound card interface is one of the most popular and capable options. For mobile operation, a radio with a built-in TNC (such as the Kenwood TM-D710) eliminates the need for any external hardware.

2.5 APRS Frequencies by Region

APRS operates on different primary frequencies depending on region. The North American frequency (144.390 MHz) is the most widely deployed, but operators traveling internationally need to know the local frequency.

Region	Primary APRS Frequency	Notes
North America (US, Canada, Mexico)	144.390 MHz	The default for nearly all APRS software and hardware
Europe	144.800 MHz	Used across most of Europe; some countries use secondary frequencies
Australia / New Zealand	145.175 MHz	
Japan	144.640 MHz	
South America	144.390 MHz (varies)	North American frequency commonly used; verify locally
New Zealand	144.575 MHz	

 Important

When operating outside your home country, always verify the local APRS frequency before transmitting. Broadcasting on the wrong frequency wastes network capacity and confuses local operators.

Chapter 3

Packet Structure

3.1 The AX.25 Frame

Every APRS packet is an AX.25 frame. AX.25 is the link layer protocol of packet radio — it handles addressing, sequencing, and error detection. For APRS, the key fields in an AX.25 frame are the source address (your callsign), the destination address (an APRS identifier), and the digipeater addresses (your path).

When you see a raw APRS packet, it typically looks like this:

```
KE4CON-1>APRS,WIDE1-1,WIDE2-1:!4158.63N/08815.78W>146.520 Home
```

Breaking this down:

Field	Example Value	What It Means
Source callsign	KE4CON-1	The transmitting station, with SSID -1 indicating a secondary station
Destination	APRS	The APRS software destination identifier (many variations exist: APNU, APWW, etc.)
Digipeater path	WIDE1-1,WIDE2-1	The relaying instructions — how many digipeater hops this packet requests
Data type indicator	!	Exclamation mark indicates a real-time position without timestamp
APRS payload	!4158.63N/08815.78W>146.520 Home	Position, symbol, and comment — the actual information being transmitted

3.2 Anatomy of an APRS Packet

The APRS information field — everything after the colon in the raw packet — begins with a data type identifier (DTI) character that tells receiving software what kind of data follows. The most common DTI characters:

DTI Character	Packet Type	Description
! or =	Position (no timestamp)	Real-time position. ! = no message capability, = = message capable
/ or @	Position (with timestamp)	Position including a time stamp. / = no msg, @ = msg capable
:	Message	Point-to-point APRS message or bulletin

;	Object	An APRS object — a named marker at a fixed location
)	Item	A lightweight object without full AX.25 addressing
–	Weather report	Complete weather observation without position
T#	Telemetry	Analog and digital sensor values from a remote station
>	Status	A text status report without position
?	Query	A request for information from other stations

3.3 Position Packets

Position packets are the most common type on the APRS network — typically 70–90% of all traffic. They report a station's location using one of several encoding formats.

Uncompressed Position

The most readable format, encoding latitude and longitude as degrees and decimal minutes:

```
!4158.63N/08815.78W>
```

Reading this: 41 degrees 58.63 minutes North, 088 degrees 15.78 minutes West. The / between the coordinates is the symbol table identifier (/ = primary table). The > following the coordinates is the symbol code (> = car/vehicle).

Compressed Position

A denser 4-byte encoding of coordinates that takes less channel bandwidth. The same position in compressed format appears as a sequence of printable ASCII characters. Compressed format can also encode altitude and radio range in the same packet. Most modern APRS software can decode both formats automatically.

Position Ambiguity

APRS supports intentional position ambiguity — reporting a position less precisely than the equipment can determine. A station might report to the nearest mile rather than the nearest hundred feet for privacy or because precise location is not needed for the operational picture. Ambiguity is indicated by replacing trailing digits in the coordinate with spaces.

3.4 Message Packets

APRS messages are point-to-point text transmissions addressed to a specific callsign. Unlike position packets which are pure broadcasts, message packets include an acknowledgment mechanism: the recipient is expected to send an ACK (acknowledgment) packet when the message is received. The sender retransmits the message until an ACK is received or the retry limit is reached.

A message packet looks like:

```
KE4CON-1>APRS,WIDE1-1::W9ABC-7 :En route, ETA 30 min{001
```

The double colon after APRS indicates a message. The addressee (W9ABC-7) is padded to 9 characters with spaces. The message text follows another colon. The {001 at the end is the message number used for acknowledgment tracking.

APRS messages have a 67-character limit for the message text. This is a protocol constraint from the original AX.25 information field limits.

Important

APRS messages are NOT private. They are transmitted over the shared APRS channel and forwarded by iGates to the global APRS-IS network, where they are visible to anyone monitoring. Never send sensitive information via APRS messages.

3.5 Weather Packets

Weather packets carry meteorological observations in a standardized format. An APRS weather station transmits temperature, wind speed, wind direction, barometric pressure, humidity, and rainfall. The format is compact and designed to be transmitted frequently.

A typical weather packet:

```
KE4CON-13>APRS,WIDE1-1:@221553z3957.68N/08259.93W_270/005g008t072r000b10152h65
```

The _ character after the coordinates is the weather station symbol. The data fields that follow encode wind direction (270°), wind speed (5 mph), gust (8 mph), temperature (72°F), rainfall (0), barometric pressure (1015.2 mb), and humidity (65%).

Field	Format	Example	Meaning
Wind direction	cDDD	c270	270 degrees (westerly wind)
Wind speed	sDDD	s005	5 mph sustained
Wind gust	gDDD	g008	8 mph peak gust in last 5 minutes
Temperature	tDDD	t072	72°F
Rainfall (1 hr)	rDDD	r000	0 hundredths of inch
Barometric pressure	bDDDDD	b10152	1015.2 millibars
Humidity	hDD	h65	65%

3.6 Object and Item Packets

Objects and items allow any station to place named markers on the APRS map on behalf of anything — not just their own radio station. Objects are used extensively in emergency communications and public service events to mark:

- Incident command post locations
- Staging areas and resource locations
- Shelter-in-place boundaries
- Repeater sites with frequency and tone information
- Net control station locations during activations

- Points of interest for event participants

An object packet:

```
KE4CON-1>APRS,WIDE1-1;;STAGING *221553z4158.63N/08815.78W!Staging Area A
```

The semicolon DTI identifies this as an object packet. The object name (STAGING) is exactly 9 characters, padded with spaces. The asterisk (*) means the object is "live" — a slash would indicate a killed (removed) object. The remaining fields are identical to a position packet: timestamp, coordinates, symbol, and comment.

3.7 Status Packets

Status packets carry a brief text announcement from a station — what the operator is doing, their availability, or relevant operational information. Unlike position packets, status packets contain no coordinates. They appear on APRS displays as a text annotation alongside the station's last known map position.

Common status packet uses: "Net control for W9NET 146.520", "QRT for the evening", "Available for EMCOMM assignment", "On scene at Elm and Main".

3.8 Telemetry Packets

Telemetry packets carry readings from sensors and instruments — up to five analog values and eight digital bit values per packet. They are used by automated stations such as weather sensors, battery monitors, repeater controllers, and remote sensing equipment.

The meaning of each analog channel is defined by companion PARM (parameter name), UNIT, and EQNS (equation) packets sent by the same station. Without these companion packets, the raw telemetry values are numbers without context.

3.9 Invalid and Malformed Packets

On any active APRS network, a small percentage of packets will be invalid or malformed. This is normal and expected. Causes include:

- RF interference corrupting the audio tones during reception
- Packet collisions when two stations transmit simultaneously
- Misconfigured or buggy APRS software at the transmitting station
- Non-standard APRS implementations or experimental packet types

A healthy APRS network typically shows 1–5% invalid packets. Higher rates (above 10–15%) may indicate a local interference problem or a nearby misconfigured station.

Chapter 4

Digipeaters and Paths

4.1 What a Digipeater Does

A digipeater (digital repeater) receives an APRS packet and re-transmits it on the same frequency. Unlike a voice repeater which repeats everything it hears, a digipeater is selective — it only retransmits packets that request relaying using specific digipeater path aliases.

When a digipeater retransmits your packet, it adds its own callsign to the path and decrements the hop counter. A packet with WIDE2-2 in its path becomes WIDE2-1 after one digipeater, and is not relayed further after a second one. This limits how far a packet can travel and prevents it from being retransmitted forever.

A well-placed digipeater — typically on a hilltop or tall building with a good omnidirectional antenna — can receive mobile and portable stations that are otherwise limited by terrain and relay their packets to a much wider audience.

 Note

Digipeaters listen on the APRS channel and transmit on the same frequency. Every digipeater retransmission occupies the shared channel. A packet with a two-hop path (WIDE1-1,WIDE2-1) may be re-transmitted by dozens of digipeaters in a well-covered area, each occupying the channel briefly. Path length directly affects network congestion.

4.2 The WIDE Path System

Modern APRS uses a "New-N paradigm" path system based on two aliases: WIDE1 and WIDE2. Each alias has a numeric suffix that counts down as the packet is relayed.

Path Element	Meaning	Typical Coverage
WIDE1-1	Request one relay from a fill-in (low-level) digipeater	Local area, low-level digipeaters only
WIDE2-1	Request one relay from a wide-area (high-level) digipeater	Regional area, hilltop and high-gain digipeaters
WIDE2-2	Request two relays from wide-area digipeaters	Two-hop wide-area coverage; more congestion
WIDE1-1,WIDE2-1	Standard two-hop path	Standard: fill-in then wide-area. Maximum recommended for most use
WIDE1-1,WIDE2-2	Three-hop path	Excessive for most use; avoid except in low-coverage areas

4.3 Choosing the Right Path

Path selection is one of the most important operating decisions in APRS. A path that is too long creates unnecessary congestion. A path that is too short may not reach the network coverage you need. The right path depends on your operating situation.

Situation	Recommended Path	Reason
Home station, good local coverage	WIDE1-1,WIDE2-1	Standard path for most fixed stations
Mobile in urban area	WIDE1-1,WIDE2-1	Standard path; fill-ins help in built-up areas
Mobile in rural area with no fill-ins	WIDE2-1 or WIDE2-2	Skip fill-in hop where none exist
Hilltop portable (SOTA/POTA)	WIDE1-1 only	Single hop; hilltop signal is already strong
Bench testing	(none — no path)	No digipeater relaying; local only
Testing in-car / driveway	WIDE1-1	Single hop; verify packet reaches a digipeater

4.4 Fill-In Digipeaters (WIDE1)

WIDE1 digipeaters (also called fill-in digipeaters) are low-level stations typically located at ground level or at modest elevations. They are designed to pick up mobile and portable stations that cannot reach a hilltop wide-area digipeater directly, and relay those packets upward to the wide-area network.

Fill-in digipeaters only respond to WIDE1-1 in the path. They do not relay WIDE2 traffic. This layered approach keeps fill-in digipeaters from amplifying already-relayed packets and creating loops.

Many home stations run as fill-in digipeaters. If you have a well-located home antenna and iGate, adding fill-in digipeater capability to your station is a valuable contribution to the local APRS network.

4.5 Wide-Area Digipeaters (WIDE2)

WIDE2 digipeaters (also called backbone or high-level digipeaters) are typically located on hilltops, towers, or tall buildings with wide-coverage antennas. They relay traffic with WIDE2-1 or WIDE2-2 in the remaining path. A well-sited WIDE2 digipeater can cover hundreds of square miles.

WIDE2 digipeaters are usually run by amateur radio clubs, emergency communications groups, or individual operators with excellent antenna sites. They form the backbone of regional APRS coverage.

4.6 Path Abuse and Network Congestion

The APRS channel has a finite capacity. In a well-covered area, every unnecessary transmission makes the channel busier and increases the packet collision rate. Several common path mistakes cause significant congestion:

- Using WIDE2-2 from a hilltop — your signal already reaches dozens of digipeaters. Requesting two relays from a high-altitude site floods the network.
- Using WIDE1-1,WIDE2-2 (three-hop path) — in most areas, two hops provide more than sufficient coverage. Three hops create excessive network load.
- Beacon intervals that are too short — mobile stations should generally beacon every 1–2 minutes while moving, or use SmartBeaconing. Fixed stations need not beacon more than every 10–30 minutes.
- Running a digipeater with RELAY or TRACE aliases — these are legacy aliases that are deprecated and should not be used. They cause unbounded relaying in areas that still respond to them.

Important

From a hilltop SOTA or POTA activation, use WIDE1-1 (single hop) only. Your signal from elevation reaches wide-area digipeaters directly. Using WIDE1-1,WIDE2-1 from a summit creates a flood of relayed packets across the entire region.

4.7 The Aloha Circle

The Aloha circle is a concept developed by Bob Bruninga WB4APR to describe the effective communications range on the shared APRS channel under a given traffic load.

On a shared channel, packet collisions increase as more stations transmit. As the channel load increases, the probability of any given packet being received successfully decreases. The Aloha circle defines the radius within which a station can reliably receive packets — as the channel gets busier, this effective radius shrinks.

The theoretical channel capacity of the 1200 baud APRS channel is approximately 18,000 bits per second. Each average APRS position packet (including AX.25 overhead and preamble) occupies approximately 0.3–0.5 seconds of channel time. This limits the practical sustainable throughput to roughly 60–120 unique station beacons per minute across the entire coverage area of a digipeater.

In dense urban areas — a major metro area during rush hour, or a large public service event — the APRS channel can become saturated. This is why beacon interval management and appropriate path selection are not just courtesy but technical necessities.

Tip

APRS Command's Packet Statistics Dashboard includes an Aloha circle calculator that estimates your effective communication range based on the observed packets-per-minute rate in your area.

Chapter 5

APRS-IS: The Internet Backbone

5.1 What APRS-IS Is

APRS-IS (the APRS Internet Service) is a global network of servers that carry APRS data over the internet. It extends the reach of the local RF network to worldwide scope and allows stations to receive and transmit APRS data without a radio.

When you connect APRS Command to APRS-IS, you receive a live stream of every APRS packet heard by iGate stations worldwide within your configured filter area. You can see stations from across your region, across the country, or anywhere in the world depending on your filter settings.

5.2 How APRS-IS Works

APRS-IS consists of a network of core servers that exchange data with each other and a large number of client servers that APRS software connects to. The core servers replicate all received packets across the network in near real-time.

Data enters APRS-IS two ways:

- From iGate stations that receive RF packets and forward them to the internet
- From software clients (like APRS Command) that inject packets directly when transmitting

Data exits APRS-IS two ways:

- To software clients receiving the stream (software-only monitoring)
- To iGates that inject internet packets back onto the local RF network (bidirectional iGates)

5.3 iGates: Bridging RF and Internet

An iGate (internet gateway) is an APRS station that bridges the local RF network and APRS-IS. iGates are the critical infrastructure that make APRS-IS useful.

iGate Type	What It Does
Receive-only (Rx) iGate	Listens on the APRS frequency, forwards received RF packets to APRS-IS. The most common type. Requires a receiver, TNC or software modem, and internet connection.
Transmit-capable (Tx) iGate (bidirectional)	Both receives and forwards RF→internet AND injects APRS-IS packets onto the local RF network. Allows APRS messages from internet-only stations to reach RF stations.

Running an iGate with a good antenna is one of the most valuable contributions an amateur radio operator can make to the local APRS network. Every iGate extends the reach of the network and improves coverage for mobile and portable stations in the area.

 Tip

A receive-only iGate can be built with a Raspberry Pi, a \$20 RTL-SDR receiver, Direwolf software, and a simple antenna. The startup cost is very low and the contribution to the local network is significant.

5.4 Server Filters

When you connect to APRS-IS, the server can send you all packets globally — which would be an enormous data stream — or a filtered subset based on your requirements. Filter strings tell the server what to send you.

The most common filter is a geographic radius filter:

```
r/41.9756/-88.4553/150
```

This filter requests all packets within 150 kilometers of the coordinates 41.9756°N, 88.4553°W. The distance in the APRS-IS filter string is always in kilometers, regardless of the unit preference in your software.

Filter Type	Syntax Example	What It Does
Range filter (most common)	r/lat/lon/dist	All packets within dist km of the specified coordinates
Prefix filter	p/KE4/W9	All packets from callsigns starting with KE4 or W9
Object filter	o/STAGING	All object packets named STAGING
Type filter	t/wn	Only weather (w) and NWS alert (n) packets
Friend filter	f/W9ABC/50	Packets from W9ABC and any station within 50 km of W9ABC
Entry station filter	e/W9XYZ-10	Packets that entered APRS-IS through iGate W9XYZ-10

5.5 Passcodes and Authentication

To transmit through APRS-IS (as opposed to just receiving), a station must authenticate with a passcode. The passcode is a number derived from the station's callsign using a well-known algorithm. It is not a security feature in the cryptographic sense — its purpose is to ensure that only licensed amateur radio operators can inject packets into the APRS-IS network, by making it computable only from the actual callsign.

Your passcode never changes. If you know your callsign, you can always compute your passcode. The algorithm is publicly documented and implemented in every APRS software package. Find your passcode at apps.magicbug.co.uk/passcode.

 Important

Keep your APRS-IS passcode private. Anyone who has your passcode can transmit packets using your callsign on APRS-IS. This would be an improper operating practice (and potentially illegal under FCC regulations regarding identification).

To receive APRS-IS data without transmitting, use -1 as your passcode. The server allows connections with -1 but marks them as read-only. All modern APRS clients support this mode.

5.6 The Global APRS-IS Network

APRS-IS carries traffic from hundreds of thousands of iGate stations worldwide. The data is publicly accessible — tools like aprs.fi allow anyone to track any APRS station globally in real time through a web browser.

Key APRS-IS servers for client connections:

Server	Port	Purpose
rotate.aprs2.net	14580	Automatically selects the best available server. Use this for most clients.
noam.aprs2.net	14580	North America regional server pool
euro.aprs2.net	14580	Europe regional server pool
asia.aprs2.net	14580	Asia/Pacific regional server pool
aunz.aprs2.net	14580	Australia/New Zealand regional server pool

Chapter 6

HF APRS**6.1 Why HF APRS Exists**

VHF APRS is line-of-sight. Even with digipeater coverage, there are large gaps — remote wilderness areas, ocean passages, mountainous terrain, and regions with no repeater infrastructure. HF APRS fills this gap by operating on shortwave frequencies that propagate over hundreds or thousands of miles without any relay infrastructure.

HF APRS is used primarily by:

- Expeditions and travelers in remote areas beyond VHF coverage
- Maritime operators crossing oceans
- Emergency communicators needing long-haul packet capability
- Balloons and high-altitude platforms (where HF propagation provides global coverage)

The tradeoff is significant: HF propagation is complex and variable, HF data rates are lower, and HF antennas are much larger than VHF antennas. HF APRS is a specialized tool for specific situations, not a replacement for VHF.

6.2 HF APRS Frequencies

Band	Frequency	Notes
80 meters	3.632.5 MHz (USB)	Regional; useful for Eastern US night coverage
40 meters	10.151.5 MHz (USB)	Most commonly used HF APRS frequency. Good continental coverage.
30 meters	10.151.5 MHz (USB)	Shared with 40m in some regions; check local band plans
20 meters	14.103.0 MHz (USB)	Good worldwide daytime propagation
17 meters	18.103.0 MHz (USB)	Less commonly used
15 meters	21.112.0 MHz (USB)	Good daytime propagation when open
10 meters	28.113.0 MHz (USB)	Excellent range when the band is open; unreliable otherwise

 Note

HF APRS frequencies are Upper Sideband (USB). The audio tones are generated the same way as VHF AFSK but transmitted via SSB instead of FM. The radio should be set to USB with the APRS audio injected at the correct audio offset to produce the desired RF frequency.

6.3 How HF APRS Differs from VHF

While the underlying APRS protocol is the same, HF APRS operation differs significantly from VHF:

Aspect	VHF APRS	HF APRS
Modulation	Bell 202 AFSK on FM	Bell 202 AFSK on USB SSB
Data rate	1200 baud	300 baud (most common) or 1200 baud
Range (no relay)	15–50 miles	Hundreds to thousands of miles depending on band
Relay infrastructure	Nationwide digipeater network	Very limited; mostly direct station-to-station
Frequency stability	VHF FM is tolerant of drift	SSB requires stable reference; drifting radios cause decode failures
Channel sharing	One national frequency	Multiple frequencies; regional variation
Packet collision model	Many stations, shared channel	Fewer stations; collisions less common but propagation can cause distant collisions

The 300 baud data rate used on most HF APRS frequencies means packets take longer to transmit. A position packet that takes 0.3 seconds at 1200 baud takes 1.2 seconds at 300 baud. This makes the effective channel capacity much lower on HF, which is why HF APRS is used far less densely than VHF.

6.4 Robust Packet Radio (Winlink Context)

While not strictly part of the APRS protocol, Winlink (a radio email system) operates on HF using Robust Packet Radio (RPR) and other modes on similar frequencies. Many operators who deploy HF APRS are also familiar with Winlink. The key difference:

- APRS is a broadcast system — all stations share a frequency and packets are sent without a specific recipient (except for APRS messages)
- Winlink is a store-and-forward email system using node-to-node connections

Both use HF packet radio technology but for fundamentally different purposes. APRS Command includes a Winlink RMS gateway locator to help operators find nearby Winlink stations.

6.5 HF Path Conventions

HF APRS path conventions differ from VHF. Because there is no HF digipeater network analogous to the VHF WIDE system, most HF APRS stations use very short or empty paths.

Path	Use Case
(no path / empty)	Direct station-to-station; no digipeating. Most common for HF.
WIDE1-1	Requests one relay from an HF gateway if available. Rarely useful.

GATE	Requests that an iGate forward the packet to APRS-IS. Used by mobile HF stations to get their position onto the internet-based map.
RFONLY	Prevents the packet from being forwarded to APRS-IS. Used when you only want RF coverage.

6.6 Practical HF APRS Setup

Setting up HF APRS requires:

- An HF transceiver capable of USB SSB operation on the desired frequency
- A TNC or software modem capable of 300 baud operation (Direwolf supports this; many hardware TNCs also support it)
- An HF antenna appropriate for the operating frequency
- Careful audio level adjustment — HF SSB amplifiers are linear, and overdrive causes harmonic distortion that degrades signal quality

The most critical difference from VHF setup is audio level. On FM, the audio level matters less because FM is a constant-deviation modulation. On SSB, the audio level directly affects the transmitted signal quality. Set your audio level to produce no more than 25–30% of rated power output when transmitting your AFSK tones.

Caution

Overdriving an SSB transceiver with AFSK causes splatter — interference on adjacent frequencies. Use a power meter and tune your audio level carefully. Start low and increase until packets decode reliably.

Chapter 7

Symbols, Objects, and the Map

7.1 The APRS Symbol System

Every APRS station and object is represented on the map by an icon — a small graphic that conveys at a glance what type of station it is. The symbol is encoded as two characters in every position packet: the Symbol Table Identifier and the Symbol Code.

The symbols are defined in the APRS Symbol Reference, available at aprs.org/symbols.html. Common symbols include a house (home station), a car or truck (mobile), an antenna (fixed radio station), a weather vane (weather station), and dozens of others representing emergency services, maritime, aviation, and amateur radio activities.

7.2 Primary and Alternate Symbol Tables

APRS defines two symbol tables, selected by the Symbol Table Identifier character:

- / (forward slash) — the primary symbol table, containing the most common symbols
- \ (backslash) — the alternate symbol table, containing additional symbols and overlay-capable versions of primary symbols

The alternate table allows a symbol to be "overlaid" with a single letter or number, modifying its appearance. For example, a diamond symbol from the alternate table with an overlay of "E" creates an "Emergency" marker.

Table	Identifier	Example Symbols
Primary	/	- (house), > (car), _ (weather station), r (antenna), D (digipeater), I (iGate)
Alternate	\	I (area object), E (emergency), A (Aid station), H (hospital), + (cross/shelter)

**Tip**

For most home stations, use / (primary table) and - (house symbol). For a mobile station, use / and > (car). These are the most recognizable and universally understood symbols.

7.3 Objects and Items

Objects and items allow any station to place named markers on the APRS map representing anything — not just their own station. The key distinction from a beacon is that objects are explicitly labeled with a name and can be placed at any coordinate, not just the transmitting station's location.

Objects (DTI ;) are full AX.25 packets with a 9-character name. Items (DTI ;) are a lightweight alternative with a 3–9 character name that omits some AX.25 fields for efficiency. For most practical purposes, objects are more commonly used.

An object can be "killed" by re-transmitting it with a / instead of * in the live/killed field. This removes it from the map on receiving stations. Objects should be killed when they are no longer valid — when a staging area is demobilized, when an event ends, or when a resource moves.

Important

Objects remain on the map until they expire (after a configurable period without retransmission) or are explicitly killed. If you create objects for an event, be sure to kill them afterward. Old objects cluttering the APRS map are a common problem after events.

7.4 Area Objects

Area objects are a special APRS extension that encodes a geometric shape (circle, rectangle, triangle, or line) in the comment field of an object packet. The shape is rendered as a drawn area on APRS clients that support area objects. They are used to mark geographic zones — incident perimeters, evacuation areas, coverage boundaries, and event zones.

The area object encoding format (per APRS Protocol Reference §11) is:

```
/A{Shape}{Color}{Width}/{Height}
```

The width and height are in tenths of kilometers, zero-padded to three digits. A circle with radius 5 km would encode as width=050 and height=050.

Note

Area object visual rendering depends on the receiving APRS client. Not all clients render area objects. APRS Command correctly encodes area objects per the specification.

7.5 Overlays and Compressed Formats

Beyond the basic symbol table, APRS supports:

- Symbol overlays — a single character placed on top of certain alternate table symbols to create specialized variants. Used for things like relay stations, emergency markers, and event-specific icons.
- Compressed position format — a denser encoding of coordinates, altitude, and radio range that reduces packet size and channel occupancy. Compatible with all modern APRS software.
- APRS Messaging enhancements — the message packet format supports directed queries, bulletins (broadcast to all stations), and group bulletins (to named groups).

These are advanced topics beyond the scope of this primer. The APRS Protocol Reference (available at aprs.org) is the authoritative source for the complete specification.

Chapter 8

Practical Operation

8.1 Getting on APRS for the First Time

Getting your first APRS station on the air is straightforward. The minimum requirement for internet-based APRS is:

- A valid amateur radio license
- An internet connection
- APRS Command (or any APRS client software)
- Your callsign's APRS-IS passcode (from apps.magicbug.co.uk/passcode)

Configure APRS Command with your callsign, location, and passcode. Connect to APRS-IS. Within seconds you will see stations on the map from your area. To transmit your position, enable APRS-IS beaconing and set a beacon interval of 10–20 minutes for a fixed home station.

For RF operation, you additionally need a radio on 144.390 MHz and a TNC or software modem. The APRS Command user manual covers the RF configuration in detail.

 Tip

Before transmitting, spend some time observing — connect to APRS-IS and watch what stations are active in your area. Look at what paths they use, how often they beacon, and what types of packets appear. Learning the local traffic patterns before transmitting helps you configure your station appropriately.

8.2 Beacon Intervals and Good Citizenship

One of the most common APRS mistakes is beaconing too frequently. Because all stations share a single frequency, every unnecessary transmission increases congestion for everyone in the area.

Station Type	Recommended Interval	Notes
Fixed home station	10–30 minutes	Position rarely changes; frequent beacons waste channel capacity
Mobile (moving)	1–2 minutes while moving	SmartBeaconing adjusts automatically based on speed and direction change
Mobile (stopped)	5–10 minutes	No need to beacon rapidly when stationary
Weather station	5–15 minutes	More frequent for severe weather; 15 min adequate for normal conditions
SOTA/POTA portable	Every 5–10 minutes	Alert chasers without flooding the channel; single-hop path

APRS Theory and Operations		A Primer
Event net control	5 minutes	Reasonable for events; allows field ops to see your position
Emergency activation	5 minutes	Frequent enough for situational awareness

SmartBeaconing is an algorithm supported by many APRS clients and trackers that adjusts the beacon interval dynamically based on speed and direction change. When moving fast in a straight line, beacons fire frequently to capture the path. When stationary or moving slowly, they slow down. SmartBeaconing is strongly recommended for mobile operation.

Note

A common test is to watch the raw packet log in APRS Command and check how many packets per minute are passing through your area. If you are in a busy urban area seeing 5+ packets per minute, reduce your beacon interval further.

8.3 APRS for Emergency Communications

APRS was designed from the beginning for emergency communications. Its strength in EmComm is providing a common operating picture — when everyone in the field has APRS tracking and your net control station is watching APRS Command, you can see at a glance where all your operators are.

Best practices for EmComm APRS deployment:

- Coordinate with your served agency's ICS structure — APRS is a supplementary tool, not a replacement for your incident communications plan
- Use the EMCOMM ACTIVATED session template or configure your beacon comment to identify your role
- Place objects at key locations — command post, staging areas, shelter locations, resource positions
- Use the Net Control window to track who is on the air and their status
- Start After-Action Report logging via the Events menu to automatically generate ICS-214 activity logs
- Coordinate with your ARES/RACES leadership on whether to use APRS-IS, RF, or both during the activation
- If running RF only, ensure your team has a plan for iGate coverage or understands that positions will not appear on internet-accessible maps

Tip

For large activations, consider assigning one operator specifically to APRS Command monitoring and object management. This person maintains the common operating picture while other operators handle voice traffic.

8.4 APRS for Public Service Events

Marathons, parades, bike rides, and similar public service events are ideal APRS applications. Field operators tracking a moving event can see exactly where they are relative to the course, command, and each other.

Pre-event APRS setup:

- Deploy objects for the start/finish lines, aid stations, and key reference points the day before or morning of the event
- Coordinate the APRS frequency and paths with your event communications plan

- Brief all operators on how to read APRS Command and what the map will show during the event
- Test RF coverage along the event route before the day — know where the coverage gaps are
- Designate a net control operator who will monitor APRS Command continuously during the event

The Public Service Event session template in APRS Command is pre-configured for this use case with a 5-minute beacon interval, 50-mile filter, and "Public Service" beacon comment.

8.5 APRS for SOTA and POTA

Summits on the Air (SOTA) and Parks on the Air (POTA) activators use APRS to alert chasers to their location. When a SOTA activator reaches the summit and starts APRS, chasers monitoring APRS Command or aprs.fi can see exactly where the activator is.

SOTA/POTA APRS configuration:

- Use WIDE1-1 path only — from a summit, your signal reaches wide-area digipeaters directly. Two hops causes unnecessary congestion across a wide area.
- Set your beacon comment to include the summit or park reference: SOTA W4T/SU-001 or POTA K-0001
- Beacon every 5–10 minutes to keep the chasers up to date
- Use the SOTA/POTA Portable session template in APRS Command — it preconfigures these settings
- After the activation, either stop beaconing or switch back to your normal home station profile to avoid confusion

8.6 Net Operations and APRS

Weekly radio nets increasingly use APRS Command alongside voice operation. When all net participants are running APRS, the net control station can see at a glance who has checked in, where they are, and whether any stations have gone quiet.

APRS enhances net operations by:

- Providing a visual roster that complements the verbal check-in list
- Letting net control send formatted messages via the Net Control window using F1–F8 keyboard shortcuts
- Automatically generating an ICS-211 check-in list for export after the net
- Allowing stations to check in via APRS message if they cannot participate by voice

The Weekly Check-In Net session template configures APRS Command for this use case: 10-minute beacon interval, 75-mile filter, "Net Control" comment.

8.7 Weather Stations on APRS

APRS weather stations contribute valuable real-time meteorological data to the APRS network. During severe weather events, a dense network of APRS weather stations provides resolution that the National Weather Service's official observation network cannot match.

Key considerations for APRS weather stations:

- Use a separate SSID for your weather station (e.g., KE4CON-13) to distinguish it from your voice/mobile operation
- Transmit every 5–15 minutes — frequent enough to capture changing conditions, not so frequent as to clog the channel
- Calibrate your sensors against a known reference — an APRS weather station with significantly incorrect readings misleads other operators who rely on the data

- Enable NWS alert monitoring in APRS Command to correlate your observations with official alerts
- Consider participating in the CWOP (Citizens Weather Observation Program) network — APRS weather data fed to CWOP contributes to NOAA models

8.8 Common Mistakes and How to Avoid Them

Mistake	Problem It Causes	Fix
Beacon path with spaces (WIDE 1-1)	APRS-IS silently rejects the packet. App reports "Transmitted" but nothing appears on aprs.fi.	Remove all spaces from the Beacon Path field. APRS Command corrects this automatically on save.
Using WIDE2-2 from a hilltop	Floods a wide geographic area with relayed packets; congests the channel.	Use WIDE1-1 only from elevated sites.
Beacon interval under 5 minutes for a fixed station	Wastes channel capacity; your position is not changing.	Set to 10–30 minutes for fixed home stations.
Using -1 as passcode when trying to transmit	APRS-IS connection is read-only with -1; transmit will always fail.	Enter your real passcode from apps.magicbug.co.uk/passcode .
Wrong coordinates in Station Setup	All distance calculations are wrong; your filter radius is in the wrong place on Earth.	Double-check decimal degree format: negative longitude for West.
Leaving event objects on the network	Stale objects confuse operators; an ICP from last month still showing on the map is misleading.	Kill all objects when the event ends by re-transmitting with / in the live field.
Transmitting APRS messages with sensitive information	APRS messages are public on APRS-IS, visible to anyone.	Use radio voice or encrypted messaging for sensitive communications.

Appendix A — APRS Frequencies by Region

Region	Frequency	Mode	Notes
North America	144.390 MHz	FM AFSK 1200 baud	Universal for the US, Canada, and Mexico
Europe	144.800 MHz	FM AFSK 1200 baud	Used across most of Europe
UK	144.800 MHz	FM AFSK 1200 baud	Same as Europe
Australia	145.175 MHz	FM AFSK 1200 baud	Also 144.575 in some areas
New Zealand	144.575 MHz	FM AFSK 1200 baud	
Japan	144.640 MHz	FM AFSK 1200 baud	
HF (40m) Worldwide	10.151.5 MHz (USB)	SSB AFSK 300 baud	Most widely used HF APRS frequency
HF (30m) Worldwide	10.151.5 MHz (USB)	SSB AFSK 300 baud	Shared with 40m in some regions; verify locally
HF (20m) Worldwide	14.103.0 MHz (USB)	SSB AFSK 300 baud	Good daytime propagation worldwide
HF (80m) Regional	3.632.5 MHz (USB)	SSB AFSK 300 baud	Regional; Eastern US night coverage

Appendix B — Common Path Reference

Path	Use Case	Hops
WIDE1-1,WIDE2-1	Standard for most fixed and mobile stations. Fill-in then wide-area relay.	2 max
WIDE1-1	Single fill-in hop. For hilltop portable where signal already reaches wide-area digis.	1 max

WIDE2-1	Single wide-area hop. For areas without fill-in digipeaters.	1 max
WIDE2-2	Two wide-area hops. Use only where single hop is insufficient and coverage is sparse.	2 max
GATE	Request iGate forwarding to APRS-IS. Useful for HF mobile stations.	N/A
NOGATE	Prevent APRS-IS forwarding. RF-only operation.	N/A
RFOONLY	Same as NOGATE — prevents iGate forwarding.	N/A
(empty/none)	No digipeater relaying. Direct only. For bench testing.	0

 Important

Never put spaces in your beacon path. WIDE1-1,WIDE2-1 is correct. WIDE 1-1,WIDE 2-1 will be silently rejected by APRS-IS.

Appendix C — Glossary

AX.25

The amateur radio packet radio link layer protocol. APRS packets are AX.25 frames.

AFSK

Audio Frequency Shift Keying. The modulation method used by APRS — two audio tones (1200 Hz and 2200 Hz) represent binary data.

Aloha Circle

The effective communication range on the APRS channel given the current traffic load. Shrinks as the channel gets busier.

APRS-IS

The APRS Internet Service. A global network of servers carrying APRS data over the internet.

Beacon

A periodic APRS transmission from a station, most commonly a position report.

Bell 202

The modem standard used by APRS — 1200 Hz mark, 2200 Hz space, 1200 baud.

Compressed Position

A denser encoding of APRS position data that reduces packet size compared to uncompressed format.

Digipeater

A station that receives APRS packets and re-transmits them, extending the range of the original transmission.

DTI

Data Type Identifier. The first character of the APRS information field, identifying the packet type.

FCS

Frame Check Sequence. A 16-bit error detection code at the end of every AX.25 frame.

Fill-in digipeater

A low-level digipeater that responds to WIDE1-1 and relays to wide-area digipeaters. Often at ground level or modest elevation.

GATE

An APRS path element requesting that an iGate forward the packet to APRS-IS.

iGate

An APRS station that bridges the local RF network and APRS-IS (the internet backbone).

Item

A lightweight APRS object type (DTI)) with a 3–9 character name.

New-N paradigm

The modern APRS digipeater path system based on WIDE1 and WIDE2 aliases, replacing the older RELAY/WIDE system.

NOGATE

An APRS path element that prevents a packet from being forwarded to APRS-IS.

Object

An APRS marker placed at any location on behalf of anything — not necessarily the transmitting station. DTI ;.

Passcode

A number derived from an amateur radio callsign used to authenticate APRS-IS transmit connections.

Path

The digipeater relay instructions encoded in an APRS/AX.25 packet. Determines how many hops the packet will be relayed.

PHG

Power, Height, Gain. An APRS extension encoding antenna characteristics for coverage estimation.

RFONLY

An APRS path element preventing iGates from forwarding the packet to APRS-IS.

SmartBeaconing

An algorithm that adjusts beacon interval dynamically based on speed and direction change. Preferred for mobile operation.

Symbol

A two-character code (table + code) identifying the map icon for a station or object.

TNC

Terminal Node Controller. Hardware or software that converts between digital data and radio audio for packet radio.

WIDE1

The fill-in digipeater alias in the New-N paradigm. WIDE1-1 requests one fill-in relay.

WIDE2

The wide-area digipeater alias in the New-N paradigm. WIDE2-1 requests one wide-area relay.